

Improved Private Simultaneous Messages Protocols for Symmetric Functions with Universal Reconstruction

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Abstract

Private Simultaneous Messages (PSM) is a kind of secure multiparty computation with minimal interaction pattern and minimal security requirement. A PSM protocol is said to be with universal reconstruction for a given function family if the algorithm of the referee (the output party) is independent of a function to be computed and the referee cannot infer the function from a protocol execution. In a recent work by Eriguchi and Shinagawa (EUROCRYPT 2025), the authors proposed a compiler to obtain a PSM protocol for symmetric functions from PSM protocols with universal reconstruction for symmetric functions with smaller domains. The authors also constructed the latter PSM protocols with universal reconstruction, by which the former PSM protocol achieves communication complexity better than the previously known protocols. In this paper, we construct the latter PSM protocols with universal reconstruction for symmetric functions more efficiently; the communication complexity is exponentially (in the input range) smaller than the protocols by Eriguchi and Shinagawa. As a consequence, we also obtain a PSM protocol (and also an ad-hoc PSM protocol and a robust PSM protocol) for symmetric functions that is more efficient than their protocol. Technically, a main ingredient of their protocols is a linear and injective encoding of histograms for the input elements, and our improvement is realized by finding a more efficient encoding of the histograms.

Keywords: Private Simultaneous Messages, symmetric functions, universal reconstruction, encoding of histograms

1 Introduction

Secure multiparty computation (MPC) is a cryptographic technology to compute some function $f(x_1, \dots, x_n)$ from the parties' secret inputs $x_1, \dots, x_n$. Among various kinds of MPC protocols, *Private Simultaneous Messages* (PSM) protocols [11, 13] are executed in the following manner: (1) The parties are divided into n input parties and a referee; (2) each (say, the i -th) input party has the own input x_i and a common random element r shared with the other input parties (and not with the referee); (3) each input party sends a message m_i to the referee only once, which is computed from (x_i, r) ; (4) and finally, the referee computes the output y by using the received messages $m_1, \dots, m_n$. The security requirement is that the referee (who does not collude with any input party) cannot infer any information on inputs $x_1, \dots, x_n$ beyond what can be deduced from the correct output y . Hence, PSM protocols are a kind of MPC protocols with minimal interaction pattern (e.g., the protocol is non-interactive) and minimal security requirement (e.g., privacy against input parties is not considered). The non-interactive property of PSM protocols is suitable especially when it is difficult for the input parties to synchronously participate in a protocol. Despite its minimality, PSM protocols can be used to obtain various kinds of cryptographic primitives such as other kinds of MPC protocols [12, 14, 15], secret sharing schemes [4], and conditional disclosure of secrets [16].

Conversely, constructions of PSM protocols from other primitives such as private information retrieval [4] and card-based protocols [20] are also studied.

Besides the protocols for general functions, there is a direction for studies of PSM protocols for some subclass of functions, in particular for symmetric functions [3, 6, 7, 9, 10, 18]. Among them, the PSM protocol of Eriguchi and Shinagawa [10] for symmetric functions $f: \{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$ has the lowest communication complexity $O(n^{2\lfloor d/3 \rfloor + 1})$ at the present. The main ingredient of their construction is PSM protocols with *universal reconstruction* [4], which are PSM protocols with additional properties that the algorithm for output computation by the referee is independent of the function f and the referee cannot infer information on the function f . In their construction, the authors first developed a compiler obtaining a PSM protocol for symmetric functions from PSM protocols with universal reconstruction for symmetric functions with smaller domains. Then the authors instantiated the latter PSM protocols with universal reconstruction by extending the idea of PSM protocols in [18] based on sequences of quadratic residues and nonresidues. Now if more efficient instantiations of the latter PSM protocols are found, then it results in a more efficient PSM protocol for symmetric functions. We note also that the universal reconstruction property for PSM protocols is by itself important because it results in simpler implementation of the referee's algorithm and also in stronger security.

1.1 Our Contribution

In this paper, we improve the construction in [10] of PSM protocols with universal reconstruction for symmetric functions. For the case of symmetric functions $\{0, 1, \dots, d'-1\}^n \rightarrow \{0, 1\}$, our protocol has the communication complexity $O(c^{d'-1}(n-1)^{d'})$ where $c = 1/\sqrt{2} + o(1) < 1$ (Theorem 3), while the communication complexity in [10] was $O((n-1)^{d'})$, therefore the improvement is exponential in terms of the input range d' . We note that as mentioned at the end of the previous paragraph, the universal reconstruction property is of independent interest. Moreover, by using our protocol above instead of the protocol in [10] as building blocks for the construction of [10], we obtain a PSM protocol for symmetric functions $\{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$ with communication complexity $O(c^{\lfloor d/3 \rfloor} n^{2\lfloor d/3 \rfloor + 1})$ where $c = 1/\sqrt{2} + o(1) < 1$ (Corollary 1), which is again exponentially better than the complexity $O(n^{2\lfloor d/3 \rfloor + 1})$ in [10].

We note that in [10], special kinds of PSM called *ad-hoc PSM* [5] and *robust PSM* [3] are also considered, and the authors also constructed such PSM protocols efficiently by using their main PSM protocol as a building block. Here, by using our proposed PSM protocol as an alternative building block, we obtain an ad-hoc PSM protocol and a robust PSM protocol for symmetric functions that are again exponentially more efficient than the constructions in [10].

We give a technical overview of our result. In the construction of PSM protocols with universal reconstruction for symmetric functions by [10], the authors first encoded each party's input $x_i \in \{0, 1, \dots, d'-1\}$ into a vector $x'_i = (x'_{i,0}, x'_{i,1}, \dots, x'_{i,d'-2}) \in \mathbb{Z}_{\geq 0}^{d'-1}$ given by

$$\begin{cases} x'_{i,j} = 1 \text{ for } j = x_i, \text{ and } x'_{i,j} = 0 \text{ for } j \neq x_i & (\text{if } x_i \neq d'-1) , \\ x'_{i,j} = 0 \text{ for every } j & (\text{if } x_i = d'-1) . \end{cases}$$

Then the sum $\hat{x} := x'_1 + \dots + x'_n$ of the vectors x'_i represents the histogram of inputs $(x_1, \dots, x_n)$ (note that the number of $(d'-1)$'s in the inputs can be recovered from $\hat{x}$ and the number n of input parties), therefore any symmetric function on $(x_1, \dots, x_n)$ is expressed as a function on $\hat{x}$. Secondly, the authors constructed a linear integer encoding of the histogram $\hat{x} = (\hat{x}_0, \hat{x}_1, \dots, \hat{x}_{d'-2})$ of the form $I_v(\hat{x}) = \langle v, \hat{x} \rangle = v_0 \hat{x}_0 + \dots + v_{d'-2} \hat{x}_{d'-2}$ for a coefficient vector $v = (v_0, v_1, \dots, v_{d'-2}) \in \mathbb{Z}_{\geq 0}^{d'-1}$ in a way that it is injective on the set of the possible histograms $\hat{x}$ and the maximum value, say L , of the image $I_v(\hat{x})$ is small. Now the truth table of the target function on $\hat{x}$ can be embedded into a sequence of $L+1$ bits via the encoding $I_v(\hat{x})$ of $\hat{x}$. Thirdly, the $(L+1)$ -bit sequence is embedded into the sequence of quadratic (non)residues modulo some prime p , which enables us to construct the desired PSM protocol similarly to [18]. The universal reconstruction property of the PSM protocol comes from a number-theoretical fact that *any* $(L+1)$ -bit sequence can be embedded by taking a *common* (sufficiently large) prime p [17]. Now the communication complexity is n times the bit

length of p , the latter being linear in L (not the bit length of L). Hence, to improve the communication complexity, it is crucial to find such a coefficient vector v with the value of L being smaller. And we indeed find a suitable coefficient vector v , by which the value of L is improved from $O(n^{d'-1})$ to $O(c^{d'-1}n^{d'-1})$ where $c < 1$ is as above. (Actually, the protocol of [10] used a certain trick to replace the value of n with $n - 1$, and the same trick can be applied to our construction as well.) Our construction of a vector v is recursive; from a $(d' - 3)$ -dimensional solution $v' = (v'_0, \dots, v'_{d'-4})$ we obtain a $(d' - 1)$ -dimensional solution $v = (v_0, v_1, v'_0, \dots, v'_{d'-4})$. Now if v_1 were sufficiently large and $v_0 \gg v_1$, then the injectivity condition for I_v would be easily satisfied; but the value of L becomes not desirable if v_0 and v_1 are too large. In order to decrease the value of L , we perform a careful analysis to find suitable values of v_0 and v_1 which are not too large and still make I_v injective. We hope that such a linear injective encoding I_v of histograms itself will be of independent interest.

1.2 Related Work

As for the lower bounds for communication complexity of PSM protocols, Eriguchi and Shinagawa [10] also gave a lower bound for PSM protocols with universal reconstruction; in the case of symmetric functions $\{0, 1, \dots, d - 1\}^n \rightarrow \{0, 1\}$, the communication complexity is at least $(1/d) \cdot \binom{n+d-1}{d-1}$. Our proposed PSM protocols with universal reconstruction have communication complexity closer to this lower bound than the previous protocols, but there is still room for further improvement. On the other hand, lower bounds for communication complexity of PSM protocols (without universal reconstruction) are also given for general functions [1, 2] and specific functions [8, 11, 19].

1.3 Organization of the Paper

In Section 2, we summarize notations and basic definitions used in this paper. In Section 3, we give some basic observations on properties of the injective encodings for histograms. In Section 4, we construct our new injective encoding for histograms. As applications, in Section 5, we give our efficient PSM protocols with universal reconstruction for symmetric functions and also obtain an efficient (plain, ad-hoc, or robust) PSM protocol for symmetric functions.

2 Preliminaries

2.1 Notations

Let $\mathbb{Z}_{>0}$ and $\mathbb{Z}_{\geq 0}$ denote the sets of positive integers and non-negative integers, respectively. For a real number r , let $\lfloor r \rfloor$ denote an integer nearest to r . For any $\ell \in \mathbb{Z}_{>0}$, the *weight* $\text{wt}(x)$ of a vector $x = (x_0, x_1, \dots, x_{\ell-1}) \in \mathbb{Z}_{\geq 0}^\ell$ is defined by $\text{wt}(x) := \sum_{i=0}^{\ell-1} x_i$. For $i \in \{0, 1, \dots, \ell-1\}$, let $e^{(i)} = (0, \dots, 0, 1, 0, \dots, 0) \in \mathbb{Z}_{\geq 0}^\ell$ denote the i -th unit vector, where the unique ‘1’ is at the i -th position, counted from the 0-th. Let $\mathcal{B}_\ell$ denote the set of vectors $x \in \mathbb{Z}_{\geq 0}^\ell$ with $\text{wt}(x) \leq 1$. For any $n \in \mathbb{Z}_{>0}$, let $\bar{\mathcal{B}}_{n,\ell}$ denote the set of sums $\sum_{k=0}^{n-1} x^{(k)}$ of n vectors $x^{(0)}, x^{(1)}, \dots, x^{(n-1)}$ chosen from $\mathcal{B}_\ell$. In other words, $\bar{\mathcal{B}}_{n,\ell}$ is the set of vectors $x \in \mathbb{Z}_{\geq 0}^\ell$ with $\text{wt}(x) \leq n$. Note that $\mathcal{B}_\ell$ and $\bar{\mathcal{B}}_{n,\ell}$ are denoted by $\mathcal{B}_{\ell+1}$ and $\bar{\mathcal{B}}_{n,\ell+1}$ in [10], respectively.

2.2 Private Simultaneous Messages

Definition 1. Let $n \geq 2$ be an integer, X_i and M_i be an input set and a message set for the i -th input party for $i \in \{1, 2, \dots, n\}$, respectively, R be a set of common randomness, Y be an output set for the referee, and $\mathcal{F}$ be a set of functions from $\prod_{i=1}^n X_i$ to Y . An n -party *Private Simultaneous Messages* (PSM) protocol for function family $\mathcal{F}$ is a tuple $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ of the following three algorithms:

- $\text{Gen}(f)$ for $f \in \mathcal{F}$: It is a randomized algorithm to generate an element r of R as common randomness for input parties.

- $\text{Enc}(i, f, x_i, r)$ for $i \in \{1, 2, \dots, n\}$, $f \in \mathcal{F}$, $x_i \in X_i$, and $r \in R$: It is a deterministic algorithm to compute a message $m_i \in M_i$ of the i -th input party.
- $\text{Dec}(f, m_1, \dots, m_n)$ for $f \in \mathcal{F}$ and $m_i \in M_i$: It is a deterministic algorithm to compute the output $y \in Y$ of the protocol.

We say that a PSM protocol Π is *correct* if for any $f \in \mathcal{F}$, for any output r of $\text{Gen}(f)$, for any $x_i \in X_i$ for $i \in \{1, 2, \dots, n\}$, for the output m_i of $\text{Enc}(i, f, x_i, r)$ for $i \in \{1, 2, \dots, n\}$, and for the output y of $\text{Dec}(f, m_1, \dots, m_n)$, we always have $y = f(x_1, \dots, x_n)$. We say that a PSM protocol Π is *secure* if there exists a randomized algorithm (*simulator*) $\mathcal{S}$ satisfying that for any $f \in \mathcal{F}$, for any $y \in Y$, and for any $(x_i)_{i=1}^n \in \prod_{i=1}^n X_i$ with $f(x_1, \dots, x_n) = y$, the output distribution of $\mathcal{S}(f, y)$ is identical to the probability distribution of $(\text{Enc}(i, f, x_i, r))_{i=1}^n$ where $r \leftarrow \text{Gen}(f)$.

In the setting of Definition 1, the *communication complexity* $\text{Comm}(\Pi)$ of the PSM protocol Π is defined by $\text{Comm}(\Pi) := \sum_{i=1}^n \log_2 |M_i|$. The *randomness complexity* $\text{Rand}(\Pi)$ of Π is defined by $\text{Rand}(\Pi) := \log_2 |R|$. The *computational complexity* $\text{Comp}(\Pi)$ of Π is defined by the sum of computational complexity for algorithms $\text{Gen}(f)$, $\text{Enc}(i, f, x_i, r)$ for $i = 1, 2, \dots, n$, and $\text{Dec}(f, m_1, \dots, m_n)$ for the worst choices of f , r , $x_1, \dots, x_n$, and $m_1, \dots, m_n$.

Definition 2. In the setting of Definition 1, we say that the PSM protocol Π is with *universal reconstruction* if a function $f \in \mathcal{F}$ is not given to the algorithm Dec nor to the simulator $\mathcal{S}$ as a part of input.

2.3 Coefficient Vectors of Linear Encoding

Let $v = (v_0, v_1, \dots, v_{\ell-1}) \in \mathbb{Z}_{\geq 0}^\ell$, which we call a *coefficient vector*. Given such a vector v and a parameter $n \in \mathbb{Z}_{>0}$, we consider the following mapping:

$$I_v : \overline{\mathcal{B}}_{n,\ell} \rightarrow \mathbb{Z}_{\geq 0}, \quad I_v(x) := \langle v, x \rangle = \sum_{i=0}^{\ell-1} v_i x_i$$

where $x = (x_0, x_1, \dots, x_{\ell-1}) \in \overline{\mathcal{B}}_{n,\ell}$. In [10], an efficient PSM protocol for symmetric functions is constructed by using such a mapping I_v that is injective on $\overline{\mathcal{B}}_{n,\ell}$. We say simply that a coefficient vector v is *injective* (with respect to parameter n) if I_v is injective on $\overline{\mathcal{B}}_{n,\ell}$. The PSM protocol in [10] becomes more efficient when we can find an injective coefficient vector v for which the maximum value $\max(I_v(\overline{\mathcal{B}}_{n,\ell}))$ in the image of I_v is smaller. In [10], the authors used the following injective coefficient vectors v (whose proof is postponed to the full version of that paper).

Proposition 1 ([10, Proposition 2]). *For any parameters $\ell \in \mathbb{Z}_{>0}$ and $n \in \mathbb{Z}_{>0}$, there exists an injective coefficient vector v for which $\max(I_v(\overline{\mathcal{B}}_{n,\ell})) = O(n^\ell)$.*

In this paper, we construct injective coefficient vectors v for which the order of $\max(I_v(\overline{\mathcal{B}}_{n,\ell}))$ is exponentially (in ℓ) lower than the construction of [10] (while the order with respect to n is not changed).

3 Basic Observations

In this section, we give some basic observations on injective coefficient vectors $v = (v_0, v_1, \dots, v_{\ell-1}) \in \mathbb{Z}_{\geq 0}^\ell$ (see Section 2.3 for the terminology). First, the domain $\overline{\mathcal{B}}_{n,\ell}$ of I_v is symmetric with respect to permutations of components in the vectors; therefore, v can be chosen (without loss of generality) in a way that v is weakly decreasing: $v_0 \geq v_1 \geq \dots \geq v_{\ell-1}$. Secondly, the fact that $\overline{\mathcal{B}}_{n,\ell}$ involves the zero vector $0 = (0, \dots, 0)$ and the unit vectors $e^{(i)}$ implies that each v_i should be positive (because otherwise $I_v(0) = 0 = I_v(e^{(i)})$). Similarly, the fact that $\overline{\mathcal{B}}_{n,\ell}$ involves different unit vectors $e^{(i)}$ and $e^{(j)}$ implies that each pair of v_i and v_j for $i \neq j$ should be different. Summarizing, we may assume without loss of generality that v is strictly decreasing and

with positive components: $v_0 > v_1 > \dots > v_{\ell-1} > 0$. In the rest of the paper, we deal with such coefficient vectors only.

Let $M(\ell, n)$ denote the minimum value of v_0 for injective coefficient vectors $v = (v_0, v_1, \dots, v_{\ell-1})$ with respect to parameter n . Then the minimum value of $\max(I_v(\bar{\mathcal{B}}_{n,\ell}))$ among injective coefficient vectors v is $n \cdot M(\ell, n)$. Moreover, when we consider the lexicographic order on coefficient vectors (i.e., $v \prec v'$ if the minimum index i for which $v_i \neq v'_i$ satisfies that $v_i < v'_i$), the smallest injective coefficient vector v gives the minimum value of v_0 and hence the minimum value of $\max(I_v(\bar{\mathcal{B}}_{n,\ell}))$ as well.

We consider the case of small parameters ℓ and n . When $\ell = 1$, it is obvious that $v = (1)$ is the lexicographically smallest injective coefficient vector, therefore $M(1, n) = 1$ and the minimum value of $\max(I_v(\bar{\mathcal{B}}_{n,1}))$ is n . When $\ell = 2$, we have the following property.

Proposition 2. *When $\ell = 2$, the lexicographically smallest injective coefficient vector is $v = (n+1, 1)$, therefore $M(2, n) = n+1$ and the minimum value of $\max(I_v(\bar{\mathcal{B}}_{n,2}))$ is $n(n+1)$.*

Proof. For $v = (n+1, 1)$, the range of $I_v(x)$ for each fixed value of x_0 is

$$[(n+1)x_0, (n+1)x_0 + n - x_0] \subseteq [(n+1)x_0, (n+1)(x_0 + 1) - 1]$$

which are disjoint for different values of x_0 . This implies that I_v is injective on $\bar{\mathcal{B}}_{n,2}$.

On the other hand, if $v = (v_0, v_1)$ and $v_0 \leq n$, then we have $I_v(x) = v_0 v_1 = I_v(x')$ for two different vectors $x = (v_1, 0)$ and $x' = (0, v_0)$ in $\bar{\mathcal{B}}_{n,2}$, therefore v is not injective. This implies that the choice of $v = (n+1, 1)$ is lexicographically the smallest. $\square$

On the other hand, when $n = 1$, it is obvious that $v = (\ell, \ell-1, \dots, 2, 1)$ is the lexicographically smallest injective coefficient vector, therefore $M(\ell, 1) = \ell$ and the minimum value of $\max(I_v(\bar{\mathcal{B}}_{1,\ell}))$ is ℓ . In the rest of the paper, we consider the other case $n \geq 2$.

4 A Recursive Construction of Injective Coefficient Vectors

In this section, we give a recursive construction of an injective coefficient vector as the following theorem. Recall from Section 3 that we only consider (without loss of generality) coefficient vectors with positive and strictly decreasing components.

Theorem 1. *Suppose that $\ell \geq 3$ and $n \geq 2$. Let $v' = (v'_0, v'_1, \dots, v'_{\ell-3}) \in \mathbb{Z}_{\geq 0}^{\ell-2}$ be any injective coefficient vector with respect to parameter n . Then a coefficient vector $v = (v_0, v_1, v'_0, v'_1, \dots, v'_{\ell-3}) \in \mathbb{Z}_{\geq 0}^{\ell}$ is injective with respect to parameter n if we put*

$$\begin{cases} v_0 := \frac{n^2 + n}{2} v'_0 + n + 2, v_1 := \frac{n^2 - n + 2}{2} v'_0 + n & \text{if } n \text{ is odd,} \\ v_0 := \frac{n^2 + 2n - 2}{2} v'_0 + \frac{n}{2} + 2, v_1 := \frac{n^2}{2} v'_0 + \frac{n}{2} + 1 & \text{if } n \text{ is even.} \end{cases}$$

In the following, we prove Theorem 1. Our goal is to derive a contradiction by assuming that $I_v(x) = I_v(x')$ for two vectors $x = (x_0, x_1, \dots, x_{\ell-1})$ and $x' = (x'_0, x'_1, \dots, x'_{\ell-1})$ in $\bar{\mathcal{B}}_{n,\ell}$ with $x \neq x'$. Put

$$z := \sum_{i=2}^{\ell-1} v_i x_i, z' := \sum_{i=2}^{\ell-1} v_i x'_i,$$

therefore

$$0 \leq z \leq v'_0 x_2 \text{ and } 0 \leq z' \leq v'_0 x'_2.$$

By symmetry, we may assume that $x_0 \geq x'_0$. Then by subtracting x'_0 from the components x_0 of x and x'_0 of x' , respectively, we still have two distinct vectors in $\bar{\mathcal{B}}_{n,\ell}$ with the same value of I_v . Indeed, for

$\tilde{x} := (x_0 - x'_0, x_1, \dots, x_{\ell-1})$ and $\tilde{x}' := (0, x'_1, \dots, x'_{\ell-1})$, we have $\tilde{x} \neq \tilde{x}'$ and

$$\begin{aligned} I_v(\tilde{x}) &= v_0(x_0 - x'_0) + \sum_{i=1}^{\ell-1} v_i x_i = v_0 x_0 - v_0 x'_0 + \sum_{i=1}^{\ell-1} v_i x_i \\ &= I_v(x) - v_0 x'_0 \\ &= I_v(x') - v_0 x'_0 \\ &= v_0 x'_0 + \sum_{i=1}^{\ell-1} v_i x'_i - v_0 x'_0 = \sum_{i=1}^{\ell-1} v_i x'_i = I_v(\tilde{x}') . \end{aligned}$$

Therefore, we may assume without loss of generality that $x'_0 = 0$.

First, we suppose that $x_1 \geq x'_1$. By the same reason as above, we may assume that $x'_1 = 0$. Now if $x_0 = x_1 = 0$, then we have $(x_2, \dots, x_{\ell-1}) \neq (x'_2, \dots, x'_{\ell-1})$ (since $x \neq x'$) and

$$I_v(x) = I_{v'}(x_2, \dots, x_{\ell-1}) = I_{v'}(x'_2, \dots, x'_{\ell-1}) = I_v(x') ,$$

contradicting the assumption that v' is injective. On the other hand, if $(x_0, x_1) \neq (0, 0)$, then the assumptions $x'_0 = x'_1 = 0$ and $I_v(x) = I_v(x')$ imply that

$$v_1 \leq v_0 x_0 + v_1 x_1 = I_v(x) - z = I_v(x') - z = z' - z \leq v'_0 x'_2 \leq v'_0 n ,$$

while we have (since $n \geq 2$)

$$v_1 - v'_0 n = \begin{cases} \frac{(n-1)(n-2)}{2} v'_0 + n > 0 & \text{if } n \text{ is odd} , \\ \frac{n(n-2)}{2} v'_0 + \frac{n}{2} + 1 > 0 & \text{if } n \text{ is even} , \end{cases}$$

therefore $v_1 > v'_0 n$. This is a contradiction.

Hence it suffices to consider the other case $x_1 < x'_1$. By the same reason as above, we may assume that $x_1 = 0$, therefore $x'_1 \geq 1$. Since $I_v(x) = I_v(x')$, we have

$$v_0 x_0 + z = v_1 x'_1 + z' .$$

We put

$$\Delta := v_0 - v_1 = \begin{cases} (n-1)v'_0 + 2 & \text{if } n \text{ is odd} , \\ (n-1)v'_0 + 1 & \text{if } n \text{ is even} . \end{cases}$$

Then we have

$$2v_1 - \Delta n = \begin{cases} 2v'_0 > 0 & \text{if } n \text{ is odd} , \\ nv'_0 + 2 > 0 & \text{if } n \text{ is even} , \end{cases}$$

therefore

$$\Delta n < 2v_1 . \quad (1)$$

Now we consider two cases: $z > z'$ and $z \leq z'$. For the former case $z > z'$, we have $v_0 x_0 + z - z' = v_1 x'_1$, therefore

$$0 \equiv v_0 x_0 + z - z' \equiv \Delta x_0 + z - z' \pmod{v_1} . \quad (2)$$

Now the right-hand side (RHS) of Eq. (2) satisfies that $[\text{RHS of (2)}] > 0$ (since $\Delta > 0$ and $z > z'$) and

$$[\text{RHS of (2)}] \leq \Delta x_0 + v'_0 x_2 \leq \Delta x_0 + v'_0(n - x_0) = (\Delta - v'_0)x_0 + v'_0 n . \quad (3)$$

Since $\Delta \geq v'_0$ for any choice of n , the RHS of Eq. (3) attains the maximum when $x_0 = n$. Therefore, we have

$$0 < [\text{RHS of (2)}] \leq \Delta n < 2v_1$$

by Eq. (1), while $[\text{RHS of (2)}] \equiv 0 \pmod{v_1}$ as above, therefore

$$[\text{RHS of (2)}] = v_1 .$$

Now if $x_0 \leq \lfloor n/2 \rfloor$, then by Eq. (3), we have

$$\begin{aligned} v_1 &= [\text{RHS of (2)}] \\ &\leq (\Delta - v'_0) \left\lfloor \frac{n}{2} \right\rfloor + v'_0 n \\ &= \begin{cases} ((n-2)v'_0 + 2) \frac{n-1}{2} + nv'_0 = \frac{n^2 - n + 2}{2} v'_0 + n - 1 = v_1 - 1 & \text{if } n \text{ is odd ,} \\ ((n-2)v'_0 + 1) \frac{n}{2} + nv'_0 = \frac{n^2}{2} v'_0 + \frac{n}{2} = v_1 - 1 & \text{if } n \text{ is even ,} \end{cases} \end{aligned}$$

a contradiction. On the other hand, if $x_0 \geq \lfloor n/2 \rfloor + 1$, then since $z > z'$, we have

$$\begin{aligned} v_1 &= [\text{RHS of (2)}] \\ &\geq \Delta \left(\left\lfloor \frac{n}{2} \right\rfloor + 1 \right) + 1 \\ &= \begin{cases} ((n-1)v'_0 + 2) \frac{n+1}{2} + 1 = \frac{n^2 - 1}{2} v'_0 + n + 2 > v_1 & \text{if } n \text{ is odd ,} \\ ((n-1)v'_0 + 1) \frac{n+2}{2} + 1 = \frac{n^2 + n - 2}{2} v'_0 + \frac{n}{2} + 2 > v_1 & \text{if } n \text{ is even ,} \end{cases} \end{aligned}$$

a contradiction. In any case, we have a contradiction in the current case $z > z'$.

From now, we consider the latter case $z \leq z'$. We have $v_0 x_0 = v_1 x'_1 + z' - z$, therefore

$$0 \equiv v_1 x'_1 + z' - z \equiv -\Delta x'_1 + z' - z \pmod{v_0} . \quad (4)$$

Now since $\Delta > 0$ and $x'_1 \geq 1$ (hence $x'_2 \leq n-1$), we have

$$[\text{RHS of (4)}] \leq -\Delta x'_1 + v'_0 x'_2 \leq -\Delta + v'_0 (n-1) < 0 \quad (5)$$

and since $z' \geq z$, we have

$$[\text{RHS of (4)}] \geq -\Delta x'_1 \geq -\Delta n > -2v_1 > -2v_0 \quad (6)$$

by Eq. (1). Hence we have

$$0 > [\text{RHS of (4)}] > -2v_0 ,$$

while $[\text{RHS of (4)}] \equiv 0 \pmod{v_0}$ as above, therefore

$$[\text{RHS of (4)}] = -v_0 .$$

Now if $x'_1 \leq \lfloor n/2 \rfloor + 1$, then by Eq. (6), we have

$$\begin{aligned} -v_0 &= [\text{RHS of (4)}] \\ &\geq -\Delta x'_1 \\ &\geq -\Delta \left(\left\lfloor \frac{n}{2} \right\rfloor + 1 \right) \\ &= \begin{cases} -((n-1)v'_0 + 2) \frac{n+1}{2} = -\left(\frac{n^2 - 1}{2} v'_0 + n + 1 \right) > -v_0 & \text{if } n \text{ is odd ,} \\ -((n-1)v'_0 + 1) \frac{n+2}{2} = -\left(\frac{n^2 + n - 2}{2} v'_0 + \frac{n}{2} + 1 \right) > -v_0 & \text{if } n \text{ is even ,} \end{cases} \end{aligned}$$

a contradiction. On the other hand, if $x'_1 \geq \lfloor n/2 \rfloor + 2$ (hence $n \neq 2$ since $x'_1 \leq n$), then by Eq. (5), we have

$$\begin{aligned}
-v_0 &= [\text{RHS of (4)}] \\
&\leq -\Delta x'_1 + v'_0 x'_2 \\
&\leq -\Delta x'_1 + v'_0 (n - x'_1) \\
&\leq -\Delta \left(\left\lfloor \frac{n}{2} \right\rfloor + 2 \right) + v'_0 \left(n - \left\lfloor \frac{n}{2} \right\rfloor - 2 \right) \\
&= \begin{cases} -((n-1)v'_0 + 2) \frac{n+3}{2} + \frac{n-3}{2} v'_0 = -\left(\frac{n^2+n}{2} v'_0 + n+3 \right) < -v_0 & \text{if } n \text{ is odd ,} \\ -((n-1)v'_0 + 1) \frac{n+4}{2} + \frac{n-4}{2} v'_0 = -\left(\frac{n^2+2n}{2} v'_0 + \frac{n}{2} + 2 \right) < -v_0 & \text{if } n \text{ is even ,} \end{cases}
\end{aligned}$$

a contradiction. In any case, we have a contradiction in the current case $z \leq z'$.

Hence we have a contradiction in any case. This completes the proof of Theorem 1.

Example 1. When $\ell = 3$ and $n \geq 2$, by using Theorem 1 with $v' := (1)$, we have the following injective coefficient vector with respect to parameter n :

$$v = \begin{cases} \left(\frac{n^2+3n+4}{2}, \frac{n^2+n+2}{2}, 1 \right) & \text{if } n \text{ is odd ,} \\ \left(\frac{n^2+3n+2}{2}, \frac{n^2+n+2}{2}, 1 \right) & \text{if } n \text{ is even .} \end{cases}$$

We conjecture that this is the lexicographically smallest injective coefficient vector, which is supported by our computer experiment for $n \leq 15$.

On the other hand, when $\ell = 4$ and $n = 2$, the injective coefficient vector obtained from Theorem 1 with $v' := (n+1, 1) = (3, 1)$ is $v = (12, 8, 3, 1)$, while our computer experiment shows that the lexicographically smallest injective coefficient vector is $v = (11, 8, 7, 2)$. When $\ell = 4$ and $n = 3$, the injective coefficient vector obtained from Theorem 1 with $v' := (n+1, 1) = (4, 1)$ is $v = (29, 19, 4, 1)$, while our computer experiment shows that the lexicographically smallest injective coefficient vector is $v = (23, 18, 15, 1)$. Therefore, Theorem 1 does not in general give the best result when $\ell \geq 4$.

Moreover, by solving the recurrence relation in Theorem 1 with initial values $v_0 := 1$ for $\ell = 1$ and $v_0 := n+1$ for $\ell = 2$ as in Section 3, we have the following result.

Theorem 2. Suppose that $\ell \geq 1$ and $n \geq 2$. Then there exists an injective coefficient vector $v = (v_0, v_1, \dots, v_{\ell-1}) \in \mathbb{Z}_{\geq 0}^\ell$ with respect to parameter n satisfying that

$$v_0 = \begin{cases} (1 + \gamma_n) \cdot \alpha_n^{\ell/2-1/2} - \gamma_n & \text{if } \ell \text{ is odd ,} \\ (n+1 + \gamma_n) \cdot \alpha_n^{\ell/2-1} - \gamma_n & \text{if } \ell \text{ is even ,} \end{cases}$$

where

$$\begin{cases} \alpha_n := \frac{n^2+n}{2}, \gamma_n := \frac{2n+4}{n^2+n-2} & \text{if } n \text{ is odd ,} \\ \alpha_n := \frac{n^2+2n-2}{2}, \gamma_n := \frac{n+4}{n^2+2n-4} & \text{if } n \text{ is even .} \end{cases}$$

In particular, when $n \rightarrow \infty$ and $\ell \rightarrow \infty$,

$$v_0 = O \left(\left(\frac{1}{\sqrt{2}} + o(1) \right)^\ell n^{\ell-1} \right) \text{ and } \max(I_v(\bar{\mathcal{B}}_{n,\ell})) = O \left(\left(\frac{1}{\sqrt{2}} + o(1) \right)^\ell n^\ell \right) .$$

5 Application to Efficient PSM for Symmetric Functions

5.1 Protocols with Universal Reconstruction

For any prime p and any $i \in (\mathbb{Z}/p\mathbb{Z})^\times$, let $\left(\frac{i}{p}\right)$ denote the Legendre symbol of i modulo p , i.e., we have $\left(\frac{i}{p}\right) = 1$ if i is a quadratic residue modulo p and $\left(\frac{i}{p}\right) = -1$ if i is a quadratic nonresidue modulo p . Let $\mathbf{Q}_p$ denote the sequence of quadratic characters modulo p , i.e., for each $i \in (\mathbb{Z}/p\mathbb{Z})^\times$, we have $\mathbf{Q}_p(i) = 0$ if $\left(\frac{i}{p}\right) = 1$ and $\mathbf{Q}_p(i) = 1$ if $\left(\frac{i}{p}\right) = -1$. For any $L \in \mathbb{Z}_{>0}$, the L -th Peralta prime [17] is the smallest prime p for which every L -bit string appears as a consecutive subsequence in the sequence $\mathbf{Q}_p$. It was shown in [17] that for any $L \in \mathbb{Z}_{>0}$, there exists the L -th Peralta prime p_L with $p_L = 2^{O(L)}$.

In the construction of the n -party PSM protocol Π for symmetric functions $\{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$ with universal reconstruction in [10, Section 5], the authors encoded an input element $x_i \in \{0, 1, \dots, d-1\}$ into a vector $x'_i \in \mathcal{B}_{d-1}$ in a way that x'_i is the (x_i) -th unit vector $e^{(x_i)} \in \mathbb{Z}_{\geq 0}^{d-1}$ if $0 \leq x_i \leq d-2$ and x'_i is the zero vector $(0, \dots, 0) \in \mathbb{Z}_{\geq 0}^{d-1}$ if $x_i = d-1$. Then it is shown in [10, Proposition 1] that for any symmetric function $f: \{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$, there exists a function $g_f: \bar{\mathcal{B}}_{n,d-1} \rightarrow \{0, 1\}$ satisfying that

$$g_f(x'_1 + \dots + x'_n) = f(x_1, \dots, x_n)$$

For any $x_1, \dots, x_n \in \{0, 1, \dots, d-1\}$ and their encodings $x'_1, \dots, x'_n \in \mathcal{B}_{d-1}$ as above. The protocol Π is described in Figure 1 (note again that the authors used the notation $\bar{B}_{N,d}$ to mean the set $\bar{\mathcal{B}}_{N,d-1}$ in our notation). In the protocol, the authors used an injective coefficient vector $v \in \mathbb{Z}_{\geq 0}^{d-1}$ with respect to parameter $n-1$, and the L -th Peralta prime $p_L = 2^{O(L)}$ where $L := \max(I_v(\bar{\mathcal{B}}_{n-1,d-1})) + 1$. In the protocol, each party sends a message consisting of a single integer modulo p_L , therefore the communication complexity is $\text{Comm}(\Pi) = O(n \log p_L) = O(nL)$. The common randomness consists of $n+1$ integers modulo p_L , therefore the randomness complexity is $\text{Rand}(\Pi) = O((n+1) \log p_L) = O(n \log p_L) = O(nL)$. Moreover, the authors explained that the most computationally expensive part of Π is to determine the element $a \in \mathbb{Z}/p_L\mathbb{Z}$, i.e., roughly speaking, at which part of the sequence $\mathbf{Q}_{p_L}$ the function to be computed can be embedded as an L -bit substring. The brute-force search of such an L -bit substring inside the $(p_L - 1)$ -bit string $\mathbf{Q}_{p_L}$ requires computational complexity $O(Lp_L) = 2^{O(L)}$. Therefore, the computational complexity of Π is $\text{Comp}(\Pi) = 2^{O(L)}$.

In [10], the authors used an injective coefficient vector v as in Proposition 1 with $L = O((n-1)^{d-1})$, therefore their protocol Π satisfies that $\text{Comm}(\Pi) = O((n-1)^d)$, $\text{Rand}(\Pi) = O((n-1)^d)$, and $\text{Comp}(\Pi) = 2^{O((n-1)^{d-1})}$ (see [10, Theorem 1]). Now we can replace the coefficient vector v with a smaller one obtained by Theorem 2 where $L = O((1/\sqrt{2} + o(1))^{d-1}(n-1)^{d-1})$. Then we obtain a more efficient protocol as follows.

Theorem 3. *There exists an n -party PSM protocol Π for symmetric functions $\{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$ with universal reconstruction satisfying that $\text{Comm}(\Pi) = O(c^{d-1}(n-1)^d)$, $\text{Rand}(\Pi) = O(c^{d-1}(n-1)^d)$, and $\text{Comp}(\Pi) = 2^{O(c^{d-1}(n-1)^{d-1})}$, where $c = 1/\sqrt{2} + o(1) < 1$.*

5.2 Efficient Protocols

In [10, Section 6], the authors proved the following result on construction of PSM protocols for symmetric functions from PSM protocols with universal reconstruction for symmetric functions on smaller domains.

Proposition 3 ([10, Theorem 3]). *Assume that for any $d' \in \mathbb{Z}_{>0}$, there exists an n -party PSM protocol $\Pi_{d'}$ for symmetric functions $\{0, 1, \dots, d'-1\}^n \rightarrow \{0, 1\}$ with universal reconstruction. Let $d_0, d_1, d_2 \in \mathbb{Z}_{>0}$ be parameters with $d_0 \leq d_1 \leq d_2$ and $d_0 + d_1 + d_2 = d-1$. Then there exists an n -party PSM protocol Π for symmetric functions $\{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$ with*

$$\begin{aligned} \text{Comm}(\Pi) &= O(n^{d_2} \cdot \text{Comm}(\Pi_{d_2+1}) + \text{Comm}(\Pi_{d-d_0})) , \\ \text{Rand}(\Pi) &= O(n^{d_2} \cdot \text{Rand}(\Pi_{d_2+1}) + \text{Rand}(\Pi_{d-d_0})) , \\ \text{Comp}(\Pi) &= O(n^{d_2} \cdot \text{Comp}(\Pi_{d_2+1}) + \text{Comp}(\Pi_{d-d_0})) + (n+d)^{O(d)} . \end{aligned}$$

Notations:

- Let $v \in \mathbb{Z}_{\geq 0}^{d-1}$ be a vector for which the map I_v is injective on $\overline{\mathcal{B}}_{n-1,d-1}$.
- Let $L := \max(I_v(\overline{\mathcal{B}}_{n-1,d-1})) + 1$.
- Let p be the L -th Peralta prime.
- For any symmetric function $f: \{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$, let $g_f: \overline{\mathcal{B}}_{n,d-1} \rightarrow \{0, 1\}$ be a function satisfying that

$$g_f(x'_1 + \dots + x'_n) = f(x_1, \dots, x_n)$$

For any $x_1, \dots, x_n \in \{0, 1, \dots, d-1\}$ and their encodings $x'_1, \dots, x'_n \in \mathcal{B}_{d-1}$ as in the text.

Gen(f):

1. Choose $r_1, \dots, r_n \in \mathbb{Z}/p\mathbb{Z}$ uniformly at random conditioned on $\sum_{i=1}^n r_i = 0 \bmod p$.
2. Choose $s'_0 \in (\mathbb{Z}/p\mathbb{Z})^\times$ uniformly at random and set $s_0 := (s'_0)^2 \bmod p$.
3. Output $r := (s_0, r_1, \dots, r_n)$.

Enc(i, f, x_i, r): $x_i \in \{0, 1, \dots, d-1\}$ is encoded into $x'_i \in \mathcal{B}_{d-1}$ as in the text.

- For $i = 1$:
 1. Compute $T = (T(0), T(1), \dots, T(L-1)) \in \{0, 1\}^L$ satisfying that

$$T(I_v(\tilde{x})) = g_f(x'_1 + \tilde{x})$$

for every $\tilde{x} \in \overline{\mathcal{B}}_{n-1,d-1}$.

2. Find an $a \in \mathbb{Z}/p\mathbb{Z}$ satisfying that

$$\left(\frac{a+j}{p} \right) = (-1)^{T(j)}$$

for every $j \in \{0, 1, \dots, L-1\}$.

3. Output $m_1 := as_0 + r_1 \bmod p$.

- For $i \neq 1$:
 1. Output $m_i := I_v(x'_i)s_0 + r_i \bmod p$.

Dec($m_1, \dots, m_n$):

1. Compute
$$q := \left(\frac{m_1 + \dots + m_n}{p} \right) \in \{1, -1\} .$$
2. Output $y := 0$ if $q = 1$ and $y := 1$ if $q = -1$.

Figure 1: The PSM protocol with universal reconstruction for symmetric functions $f: \{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$ in [10]

In [10, Corollary 1], by using this result and their PSM protocols with universal reconstruction as building blocks, the authors derived an n -party PSM protocol Π for symmetric functions $\{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$ with $\text{Comm}(\Pi) = O(n^\delta)$, $\text{Rand}(\Pi) = O(n^\delta)$, and $\text{Comp}(\Pi) = 2^{O(n^{\delta-1})}$, where $\delta = 2\lfloor d/3 \rfloor + 1$. Now by using our PSM protocols in Theorem 3 instead, we obtain a more efficient protocol as follows.

Corollary 1. *For any $n \geq 2$ and $d \in \mathbb{Z}_{>0}$, there exists an n -party PSM protocol Π for symmetric functions $\{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$ with*

$$\text{Comm}(\Pi) = O(c^{\delta'} n^\delta), \text{Rand}(\Pi) = O(c^{\delta'} n^\delta), \text{Comp}(\Pi) = 2^{O((cn)^{\delta-1})},$$

where $c = 1/\sqrt{2} + o(1) < 1$, $\delta = 2\lfloor d/3 \rfloor + 1$, and $\delta' = \lfloor d/3 \rfloor$.

Proof. By the same argument as in the proof of Corollary 1 in [10], there exist parameters d_0, d_1, d_2 with $d_0 \leq d_1 \leq d_2$, $d_0 + d_1 + d_2 = d-1$, $\delta := \max\{2d_2+1, d-d_0\} = 2\lfloor d/3 \rfloor + 1$, and $\delta' := \min\{d_2, d-d_0-1\} = \lfloor d/3 \rfloor$. Indeed:

- If $d = 3t$ ($t \in \mathbb{Z}_{>0}$), we set $d_0 = t-1$ and $d_1 = d_2 = t$. Then $\delta = 2t+1 = 2\lfloor d/3 \rfloor + 1$ and $\delta' = t = \lfloor d/3 \rfloor$.
- If $d = 3t-1$, we set $d_0 = t-2$ and $d_1 = d_2 = t$. Then $\delta = 2t+1 = 2\lfloor d/3 \rfloor + 1$ and $\delta' = t = \lfloor d/3 \rfloor$.
- If $d = 3t-2$, we set $d_0 = d_1 = d_2 = t-1$. Then $\delta = 2t-1 = 2(t-1)+1 = 2\lfloor d/3 \rfloor + 1$ and $\delta' = t-1 = \lfloor d/3 \rfloor$.

For each $d' \in \mathbb{Z}_{>0}$, let $\Pi_{d'}$ be the PSM protocol in Theorem 3 for symmetric functions $\{0, 1, \dots, d'-1\}^n \rightarrow \{0, 1\}$, therefore $\text{Comm}(\Pi_{d'}) = O(c^{d'-1} n^{d'})$, $\text{Rand}(\Pi_{d'}) = O(c^{d'-1} n^{d'})$, and $\text{Comp}(\Pi_{d'}) = 2^{O(c^{d'-1} n^{d'-1})}$, where $c = 1/\sqrt{2} + o(1) < 1$. Now we have

$$\begin{aligned} n^{d_2} \cdot \text{Comm}(\Pi_{d_2+1}) &= O(c^{d_2} n^{2d_2+1}), \text{Comm}(\Pi_{d-d_0}) = O(c^{d-d_0-1} n^{d-d_0}), \\ n^{d_2} \cdot \text{Rand}(\Pi_{d_2+1}) &= O(c^{d_2} n^{2d_2+1}), \text{Rand}(\Pi_{d-d_0}) = O(c^{d-d_0-1} n^{d-d_0}), \\ n^{d_2} \cdot \text{Comp}(\Pi_{d_2+1}) &= 2^{O((cn)^{d_2})}, \text{Comp}(\Pi_{d-d_0}) = 2^{O((cn)^{d-d_0-1})}, \end{aligned}$$

therefore (since $d_2 \leq \delta-1$ and $c < 1$)

$$\begin{aligned} n^{d_2} \cdot \text{Comm}(\Pi_{d_2+1}) + \text{Comm}(\Pi_{d-d_0}) &= O(c^{\delta'} n^\delta), \\ n^{d_2} \cdot \text{Rand}(\Pi_{d_2+1}) + \text{Rand}(\Pi_{d-d_0}) &= O(c^{\delta'} n^\delta), \\ n^{d_2} \cdot \text{Comp}(\Pi_{d_2+1}) + \text{Comp}(\Pi_{d-d_0}) &= 2^{O((cn)^{\delta-1})}. \end{aligned}$$

Hence by Proposition 3, we have a PSM protocol Π as in the statement. $\square$

5.3 Ad-Hoc PSM

For parameters $1 \leq k \leq n$, a k -out-of- n *ad-hoc PSM* protocol for a k -input function f is a special kind of an n -party PSM protocol in which some parties may not send their messages to the referee and if some set of k parties sends their messages $m_{i_1}, \dots, m_{i_k}$, then the referee can obtain the function value $f(x_{i_1}, \dots, x_{i_k})$ on their inputs. The security requirement is that if exactly k parties send messages then the referee cannot infer any information beyond the function value, and if only less than k parties send messages then the referee cannot infer any information (here we do not consider the case where more than k parties send messages). See [5] for a formal definition of ad-hoc PSM. We note that n -out-of- n ad-hoc PSM is not the same notion as plain n -party PSM because plain PSM does not consider the situation where only less than n parties send messages.

In [10], the authors gave the following result to obtain a k -out-of- n ad-hoc PSM protocol from an n -out-of- n ad-hoc PSM protocol.

Proposition 4 ([10, Theorem 4]). *If there exists an n -out-of- n ad-hoc PSM protocol Π for symmetric functions $\{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$, then there exists a k -out-of- n ad-hoc PSM protocol Π' for symmetric functions $\{0, 1, \dots, d-1\}^k \rightarrow \{0, 1\}$ with*

$$\begin{aligned}\text{Comm}(\Pi') &= O(n \cdot \max\{\text{Comm}(\Pi), \log_2 n\}) , \\ \text{Rand}(\Pi') &= \text{Rand}(\Pi) + O(n^2 \cdot \max\{\text{Comm}(\Pi), \log_2 n\}) .\end{aligned}$$

Moreover, in the proof of [10, Corollary 3], the authors deduced the following property.

Proposition 5 ([10, Proof of Corollary 3]). *The n -party PSM protocol obtained by applying Proposition 3 to the PSM protocol in Figure 1 is an n -out-of- n ad-hoc PSM protocol. Hence, the PSM protocol in Corollary 1 is an n -out-of- n ad-hoc PSM protocol for symmetric functions $\{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$.*

By combining Propositions 4 and 5, we obtain the following protocol, which is again exponentially (in d) more efficient than the construction of [10, Corollary 3] with $\text{Comm}(\Pi) = O(n^{\delta+1})$ and $\text{Rand}(\Pi) = O(n^{\delta+2})$ where $\delta = 2\lfloor d/3 \rfloor + 1$.

Corollary 2. *There exists a k -out-of- n ad-hoc PSM protocol Π for symmetric functions $\{0, 1, \dots, d-1\}^k \rightarrow \{0, 1\}$ with*

$$\text{Comm}(\Pi) = O(c^{\delta'} n^{\delta+1}), \text{Rand}(\Pi) = O(c^{\delta'} n^{\delta+2}) ,$$

where $c = 1/\sqrt{2} + o(1) < 1$, $\delta = 2\lfloor d/3 \rfloor + 1$, and $\delta' = \lfloor d/3 \rfloor$.

5.4 Robust PSM

For parameters $0 \leq t \leq n-1$, we consider a situation where the referee colludes with a subset of t input parties; let x_H denote the collection of inputs from the honest input parties (i.e., those not colluding with the referee) and let $x_{\overline{H}}$ denote the collection of inputs from the colluded input parties. In this situation, it cannot be prevented that the referee learns more function values $f(x_H, x_{\overline{H}})$ by changing the part of inputs $x_{\overline{H}}$ and repeating the protocol with given fixed messages from the honest parties (corresponding to a given fixed x_H). We say that a PSM protocol is t -robust if the referee cannot infer any information beyond what is unavoidable as described above. See [3] for a formal definition of robust PSM.

Benhamouda, Krawczyk, and Rabin [7] proposed a compiler from a plain PSM protocol to a t -robust PSM protocol. In [10, Corollary 5], by applying the compiler to their main protocol, the authors obtained a t -robust PSM protocol Π for symmetric functions $\{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$ with $\text{Comm}(\Pi) = n^{\delta} P^{t+O(1)}$ and $\text{Rand}(\Pi) = n^{\delta} P^{t+O(1)}$, where $\delta = 2\lfloor d/3 \rfloor + 1$ and P is the smallest prime power satisfying that $P \geq \max\{n, d\}$. On the other hand, by applying the same compiler to our main protocol in Corollary 1, we obtain the following protocol, which is again exponentially (in d) more efficient than the construction of [10, Corollary 5].

Corollary 3. *There exists a t -robust n -party PSM protocol Π for symmetric functions $\{0, 1, \dots, d-1\}^n \rightarrow \{0, 1\}$ with*

$$\text{Comm}(\Pi) = c^{\delta'} n^{\delta} P^{t+O(1)}, \text{Rand}(\Pi) = c^{\delta'} n^{\delta} P^{t+O(1)} ,$$

where $c = 1/\sqrt{2} + o(1) < 1$, $\delta = 2\lfloor d/3 \rfloor + 1$, $\delta' = \lfloor d/3 \rfloor$, and P is the smallest prime power satisfying that $P \geq \max\{n, d\}$.

6 Conclusion

In this paper, we constructed PSM protocols with universal reconstruction for symmetric functions that are more efficient than the previous protocols by Eriguchi and Shinagawa [10], by finding more efficient linear and injective encoding of the histograms for input elements. As a consequence, we also obtained a PSM protocol (and also an ad-hoc PSM protocol and a robust PSM protocol) for symmetric functions that is more efficient than the previous protocol in [10]. Future research directions include further improvement for our encoding of histograms, other applications of such encoding of histograms, and other applications of PSM protocols with universal reconstruction.

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